

OFFLINE TRAINING PHASE

3DWire
Wireframes
(14,293)

Volumetric
Generation
(4-part STL)

FEA Simulation
(SfePy, ASCE 7-22)
 σ , u , C

Deep Ensemble Training
(5× Surrogate3DResNet)
11,178 samples

ONLINE OPTIMIZATION PHASE

128³ Voxel Grid
+ Part Labels

Phase 1:
Sensitivity Erosion
(99% removal)

Phase 2:
Endgame
(single voxel)

Phase 3:
Swap Refinement

Post-Process
+ STL Export
(marching cubes)

Surrogate predictions